

Experiment 5 – Bias–Variance, Overfitting and Underfitting Analysis

CO1 • Scikit-learn, NumPy, Matplotlib

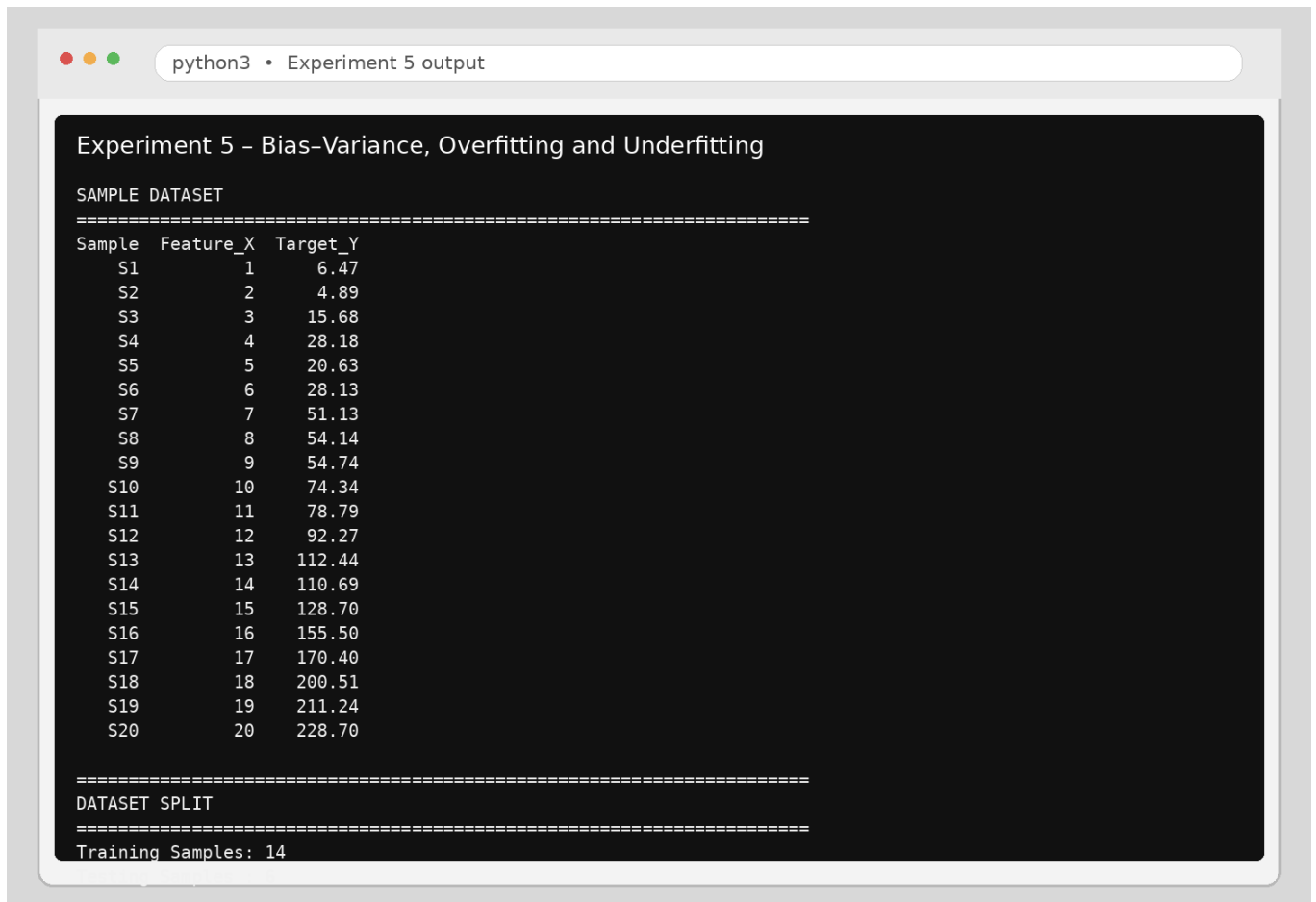


Figure 1: Sample dataset and training/testing dataset split.



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MODEL EVALUATION

Polynomial_Degree	Training_MSE	Testing_MSE	Training_RMSE	Testing_RMSE
1	208.0515	356.9260	14.4240	18.8925
2	30.8191	75.0422	5.5515	8.6627
15	23.1571	482.6390	4.8122	21.9690

BIAS-VARIANCE ANALYSIS

Degree 1: Training MSE = 208.0515, Testing MSE = 356.9260 --> UNDERFITTING - High Bias
Degree 2: Training MSE = 30.8191, Testing MSE = 75.0422 --> GOOD FIT - Balanced Bias and Variance
Degree 15: Training MSE = 23.1571, Testing MSE = 482.6390 --> OVERFITTING - High Variance

STATISTICAL EVALUATION

Minimum Testing MSE : 75.0422
Best Polynomial Degree: 2
Maximum Testing MSE : 482.6390
Worst Polynomial Degree: 15
Average Training MSE : 87.3425

Figure 2: Model evaluation, bias–variance analysis and statistical evaluation.



python3 • Experiment 5 output

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FINAL RESULT

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Bias-Variance, Overfitting and Underfitting Analysis was successfully performed using Scikit-learn, NumPy and Matplotlib.

The experiment demonstrated:

1. Sample dataset preparation
2. Training and testing dataset split
3. Polynomial regression models
4. Underfitting analysis
5. Good fit analysis
6. Overfitting analysis
7. Training MSE and Testing MSE
8. RMSE-based model evaluation
9. Bias-Variance trade-off analysis
10. Graphical comparison of model performance

The results were successfully analyzed and visualized

Figure 3: Final experiment result.

Performance Comparison



Figure 4: Training MSE and Testing MSE comparison.

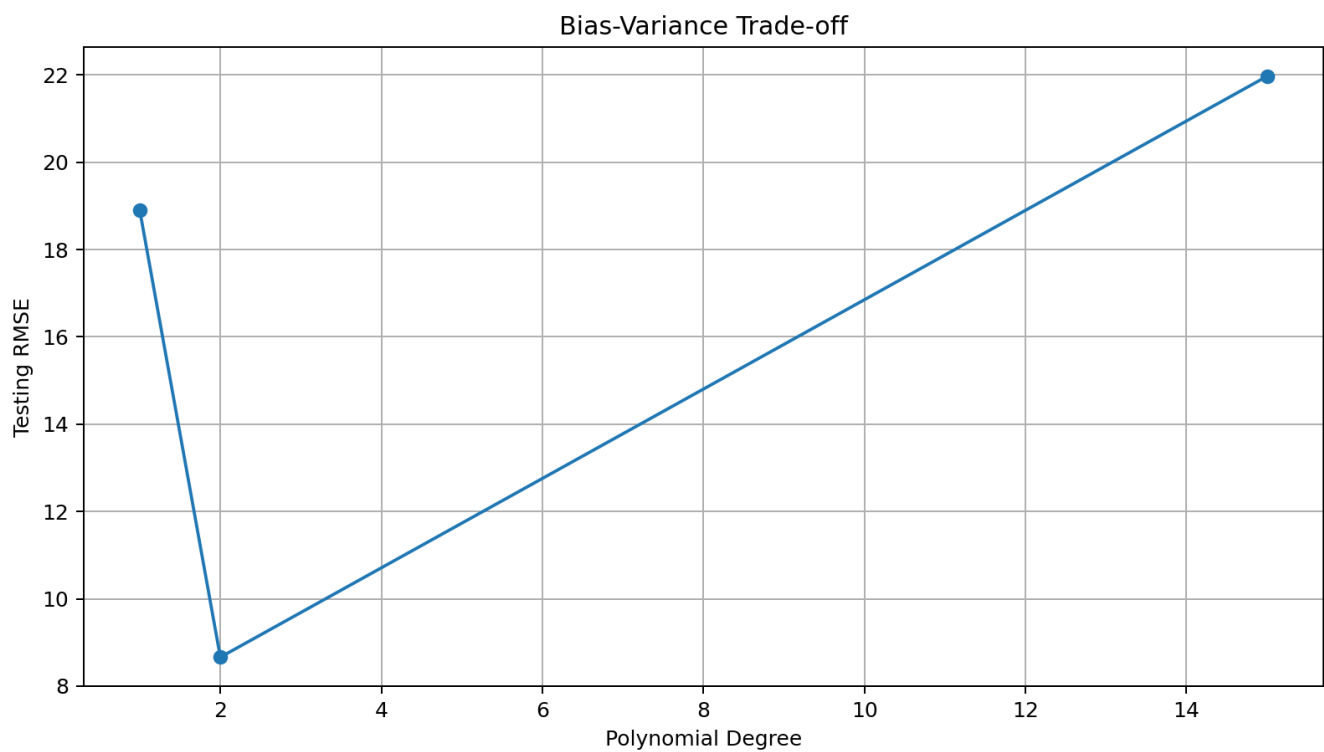


Figure 5: Testing RMSE and bias–variance trade-off.

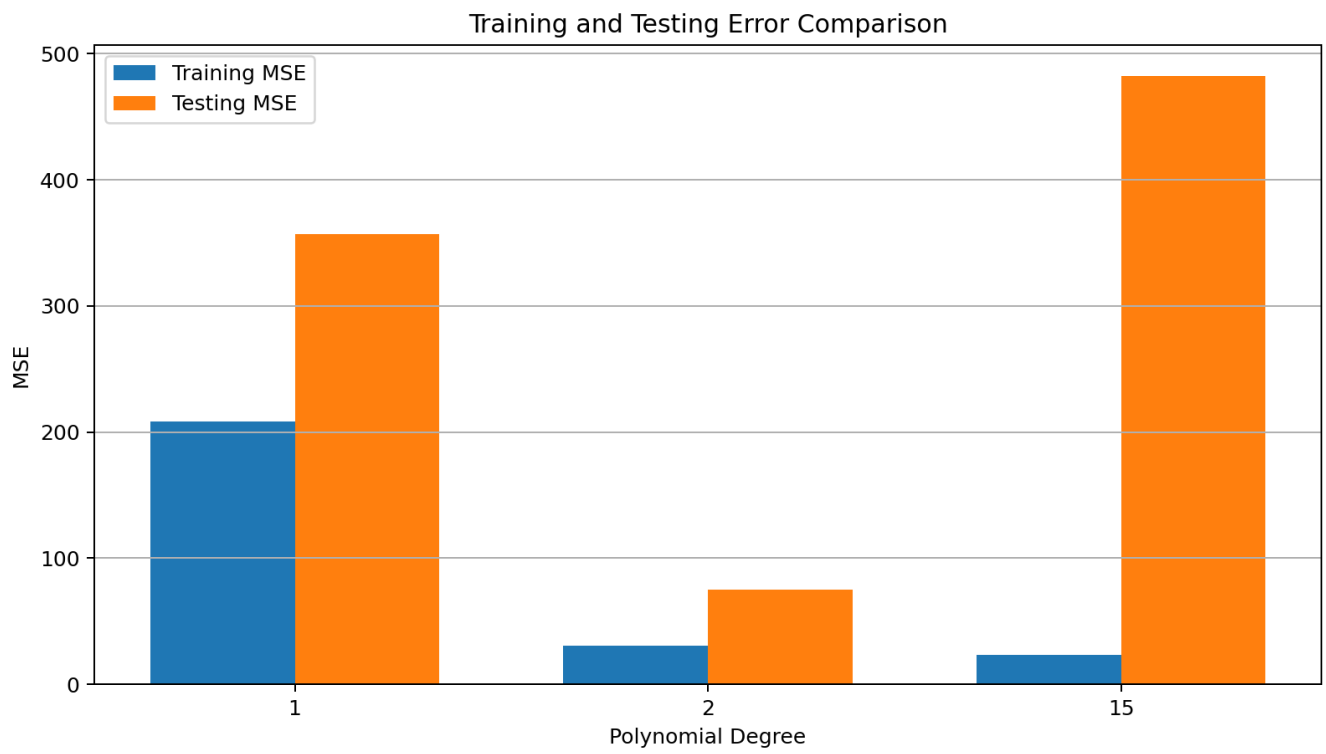


Figure 6: Training and testing error comparison.